# redBus - Marketing Analytics & Insights Lead (AI-Led)

## Interview prep pack: cases, frameworks, SQL, industry notes, resume deep dive

Prepared for Chris D'Mello. Round: first interview with Director of Marketing, then case study round. Built from our prep sessions plus the JD.

### How to use this pack

- Read Section 1 (pitch + one-page cheat sheet) the day before. Sections 4-5 are for practising cases out loud.
- Every case follows the same shape: Clarify, Framework, Steps, Recommendations, Pitfalls, Resume hook, Probes, Spoken answer. Reproduce that shape in the room; it shows depth of thinking.
- All numbers in examples are **illustrative** unless stated as yours. Do not present them as redBus or Flipkart facts.
- Items marked **[verify]** are general market knowledge not checked against current sources. Confirm before quoting.
- Section 6 (SQL) uses an **assumed schema**. Adapt table and column names. SQL was not discussed earlier in this chat in detail, so this section is new material built for the role.

### Contents


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| 1 | Pitch, bridge lines, one-page cheat sheet |
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# 1. Pitch, bridges and cheat sheet

## 30-second pitch

*I lead loyalty analytics at Flipkart, with 9 years in e-commerce. My work sits on three things: measuring true incrementality through holdouts and causal models, retention through churn and uplift targeting, and using AI to speed up insight generation. redBus is attractive because repeat and cross-sell across Bus, Hotel and Rail is the same problem I have been solving, in a new domain.*

## Longer blurb (tailored to the JD)

*Growth and loyalty analytics leader with 9+ years in e-commerce. I answer 'what should we do differently to grow better?' rather than report numbers. At Flipkart I built the 2025/26 loyalty AOP (10% Supercoins cost savings), segmented users into cohorts to find retention drivers, and built churn and conversion models for at-risk and high-value users. I use holdouts and causal methods to separate real impact from attributed impact, including a Meta incrementality test that supported scaling budgets. I use AI to cut RCA time by 50%. I want to bring this to repeat usage, CLM and cross-sell across redBus's Bus, Hotel and Rail lines.*

## Bridge for the travel-domain gap

*Bus travel is episodic, so repeat is lumpy and timing and route context matter more than campaign volume. Loyalty and e-commerce repeat taught me the mechanics: cohorts, targeting, holdouts. I would learn the route and seasonality patterns in the first 30 days.*

## JD match check (honest)

| JD requirement | Your evidence | Status |
|---|---|---|
| 8-10 yrs marketing/growth analytics | 9+ yrs | Match |
| Repeat, retention, CLM | Churn models, 100K+ retained/month, festive holdout | Match |
| AI-driven thinking | Claude skills, 50% faster RCA (on newer resume) | Match |
| Marketing spend optimization | CAC below Rs 250, Meta holdout | Match; email/comms channels thin |
| Translate data to decisions | Budget scaling, AOP savings | Match; avoid 'WBR' sounding like reporting |
| New segments / cohorts | Cohort deep dives (MPC profiling) | Partial |
| Opportunity sizing | Not on resume | Gap: rehearse Case 6 |
| Unified customer profiles / cross-sell | Netflix partnership (check relevance) | Gap |
| NPS / feedback to predict churn | Not on resume | Gap: rehearse Case 8 |
| Cohort-level predictions | Cohort personalization, units model | Partial |

## One-page cheat sheet

| Topic | Core line |
|---|---|
| Is a channel working? | Incremental, not attributed. Holdout, then incremental ROI. |
| CAC up, budget flat | Mix vs rate; cut low-incrementality spend; reallocate by marginal ROAS; judge on repeat-adjusted LTV/CAC. |
| Attribution misleads | Last-touch credits people who would buy anyway. Meta holdout showed the gap. |
| Acquisition vs retention | One portfolio; equalize marginal LTV per rupee; a leaky bucket makes acquisition costlier. |
| Brand vs performance | MMM plus geo tests; judge brand over 6-12 months; move in steps. |
| Why equalize marginal return? | If the next rupee earns more in A than B, moving money gains for free. Optimum: equal marginal returns. |
| Repeat rate fell | Real? Mix vs rate. Hypothesis tree. Waterfall. Action + holdout. |
| CLM design | Who (uplift), when (gap model, triggers), channel (cheapest that lifts), message (value first), measure (global holdout). |

| Topic | Core line |
|---|---|
| MMM honesty | Team has not run a full MMM; you run the incrementality tests that calibrate one and know the structure. |
| Causal language | Never 'the model retained them'. Say 'we targeted; the holdout showed X incremental'. |

# 2. redBus context, industry and trends

## Facts from the JD

| Item | Detail |
|---|---|
| Ownership | redBus business of Makemytrip (India) Pvt Ltd; part of GO-MMT group (Nasdaq: MMYT) |
| Scale | Up to 250,000 transactions a day; 400,000+ live routes; 6,300+ bus operators and 50,000+ routes in India; 10M+ customers in India |
| Geographies | India, Singapore, Malaysia, Indonesia, Vietnam, Cambodia, Peru, Colombia |
| Role | Lead - Marketing Analytics and Insights (AI-led); reports to CMO; team size 2; Bengaluru; 8-10 yrs |
| Mandate | 'Not a reporting role': answer 'What should we do differently to grow better?' |
| Works with | Brand, CLM, Data Science and other teams |
| Scope | New and existing users; repeat usage and retention; new segments and cohorts; opportunity sizing for all LoBs; marketing spend incl. email/comms; unified customer profiles; cross-sell journeys and comms timing; decompose repeat rates; NPS to predict churn; cohort-level predictions; Bus x Hotel x Rail insights |

## How the bus-ticketing business works (for case reasoning)

- **Two-sided marketplace:** operators (supply) and passengers (demand). Supply quality on a route drives conversion, cancellations and NPS.
- **Episodic purchase:** most users book a few times a year. Repeat is driven by trip need (commute, festival, visit home), not by browsing. Timing models beat blanket campaigns.
- **First-trip experience matters a lot:** cancellations, delays, refund time and boarding issues make or break the second booking.
- **Seasonality:** festivals (Diwali, Pongal, Eid, Durga Puja), long weekends, summer, exam and holiday periods. Compare like-for-like calendar windows.
- **Route-level economics:** corridors differ in competition (rail, flights, other apps, operator direct sales), supply depth and price.
- **Cross-business:** the group sells Hotel and Rail, so a bus booking reveals a destination and dates. Cross-sell is the strategic angle, with cannibalization risk (rail vs bus).

## Industry and trend notes [verify before quoting]

| Trend | Why it matters in an interview |
|---|---|
| Competitive set: other online bus apps, travel super-apps, rail apps, state transport apps, new branded intercity bus operators [verify names] | Shows you think about substitutes and corridor-level competition |
| Rising acquisition costs and app-install saturation across Indian consumer apps | Why repeat and CLM economics beat paid acquisition at the margin |
| Privacy and platform changes weakening user-level tracking and attribution | Why incrementality tests and MMM are being revived |
| WhatsApp and RCS as primary CLM channels; push opt-in fatigue | Channel economics: cost per message vs lift, frequency caps |
| Growth of tier-2 and tier-3 city travel demand | New segment and opportunity-sizing angle |
| Generative AI in analytics: automated RCA, SQL generation, LLM feedback classification, insight digests | The 'AI-led' part of the title; you have real examples (Claude skills) |
| Super-app and cross-sell strategies in travel | Bus x Hotel x Rail unified profile and journey mapping |

Prep action: read redBus's recent press and the MMT investor commentary the night before, and add two current facts to your 'why redBus' answer.

# 3. Marketing instruments, Flipkart numbers, MMM and optimization

## 3.1 Instruments

### Organic / owned (no per-click spend)

| Instrument | What it is | Example |
|---|---|---|
| SEO | Ranking in unpaid search through content, site speed, links | 'Bangalore to Chennai bus tickets' on page 1 with no CPC. Has a cost (content, tech) but is not a media spend line |
| Direct / organic app | Users open the app or type the URL | Repeat users |
| CRM / CLM | Email, push, WhatsApp, SMS to existing users | Return-leg push |
| Referral | Users bring users | Refer-and-earn |

### Paid performance (action-based)

| Instrument | What it is | Example |
|---|---|---|
| SEM (paid search) | Bid on keywords, pay per click. Split brand (searching 'redbus', high intent) vs non-brand (generic, new reach) | Ad on 'bus tickets Bangalore Chennai' |
| Paid social | Meta/Instagram etc. targeted by audience | App-install or festive campaign |
| App-install campaigns | Optimized for installs (Google UAC, Meta) | New-user acquisition |
| Display / programmatic | Banners and video on third-party inventory | Retargeting after a route search |
| Affiliates / cashback | Partners paid per sale | Coupon or cashback sites |

### Paid brand (reach-based, slow payoff)

- TV / OTT / CTV (e.g. cricket tournament spots)
- Outdoor / OOH (hoardings, bus-stand branding - relevant for redBus)
- Influencer / creator content
- Radio, print, regional campaigns
- Sponsorships and events

**SEO vs SEM: SEO = organic. SEM = paid search (sometimes used for both; clarify in the room).**

## 3.2 Flipkart numbers you gave me

| Item | Value | Note |
|---|---|---|
| Monthly performance marketing spend | Rs 50 Cr | Based on customer actions (clicks, add-to-cart, buy-now) |
| Last-touch attributed revenue | Rs 600 Cr per month | Attributed, not incremental |
| ROAS | About 10-12x | 600/50 = 12x; you quoted about 10x. Use your figure |

Key framing: this ROAS is attributed. Brand SEM and retargeting capture users who would have bought anyway, so incremental ROAS is lower. Illustration (made-up): if a holdout shows only 40% of attributed revenue is truly incremental, incremental ROAS is about 4-5x, not 10-12x. Do not present this as a real Flipkart result.

## 3.3 How the instruments enter an MMM

*Sales(t) = baseline + sum over channels of beta(i) x saturation( adstock( spend(i,t) ) ) + controls + error*

- **Media variables:** spend or impressions per instrument (SEM brand, SEM non-brand, Meta, display, TV, OOH, influencer).
- **Adstock (carryover):** effects last beyond the spend week. TV decays slowly, search almost immediately. Note: if your 'alpha' means adstock decay it controls timing, not the budget split.

- **Saturation:** diminishing returns; each extra rupee earns less.
- **Controls:** seasonality, festivals and sale events, price and discounts, SEO / organic trend, competitor activity, macro.
- **Pitfall:** brand SEM spend rises with demand (people search because they already want to buy), so MMM can over-credit it. Fix with controls, priors, and calibration against holdout tests.
- **Honest line:** your team has not run a full MMM. Say you run the incrementality tests that calibrate it and understand the structure.

## 3.4 Budget optimization: why equalize marginal return

### Swap argument (intuition)

If the last rupee in Channel A earns Rs 3 and in Channel B earns Rs 1, moving Rs 1 from B to A loses 1 and gains 3 at the same total budget. The split cannot be optimal. Keep swapping until no swap helps: that is exactly when the last rupee earns the same everywhere.

### Numeric example (illustrative)

Budget Rs 100, two identical channels, revenue = 10 x sqrt(spend). Split 90/10: revenue 94.9 + 31.6 = 126.5; marginal return on A = 5/sqrt(90) = 0.53, on B = 5/sqrt(10) = 1.58. Split 50/50: 70.7 + 70.7 = 141.4, marginals equal. Moving money to B raised total revenue until marginals matched.

### Why marginal, not average

Average ROAS says how the whole spend did, including early cheap wins. Marginal ROAS says what the next rupee does, which is the only thing a budget decision changes. A saturated channel can have a great average and a poor marginal return.

### The math and the correction to the 'ratio of weights' idea

Assume Revenue(i) = beta(i) x spend(i)^alpha(i), with 0 < alpha < 1 as the saturation exponent. Maximize total revenue subject to total spend = Budget. Condition: beta(i) x alpha(i) x spend(i)^(alpha(i)-1) = lambda for every channel, where lambda is the value of one extra rupee of budget.

| Response shape | Optimal split |
|---|---|
| Log response: beta x ln(spend) | Exactly proportional to beta |
| Power response, same alpha for all channels | Proportional to beta^(1/(1-alpha)), not to beta itself |
| Different alpha per channel | More saturated channels (lower alpha) get less than their beta share |
| Linear (no saturation) | Corner solution: everything to the highest beta |

**Correction: 'split in the ratio of the weights' is right only for the log case. In general, solve for equal marginal returns. (An earlier note in our chat said the beta ratio holds for alpha = 0.5; that was wrong. At alpha = 0.5 the split is proportional to beta squared.)**

### Practical method

- Compute marginal ROAS per channel from fitted curves.
- Shift budget from lowest to highest marginal ROAS in small steps; stop when they converge.
- Tooling: scipy.optimize.minimize (SLSQP) with bounds and a sum constraint.
- Constraints: channel caps and floors, brand floor, payback period, repeat-adjusted value instead of first-booking revenue.
- Uncertainty: optimize on scenarios (low/base/high) or Bayesian posteriors; prefer robust shifts; validate large moves with a geo test.

> *From the MMM I get each channel's response curve. I would not split budget by the coefficient ratio because that ignores saturation. I would allocate so marginal return per rupee is equal across channels, then add real constraints and validate the shift with a geo test.*

# Case 1: Repeat rate fell 3 points YoY

*Prompt: Repeat rate fell 3 pts YoY. Diagnose.*

## Step 0 - Clarify (say these out loud)

- Definition: repeat = 2nd booking within 90 days (cohort) or % of this month's bookers who are repeat (stock)? They move differently.
- Shape: gradual or step-change? Which month did it start?
- Scope: all LoBs and geographies or one?
- Decision it informs: fix CLM, shift acquisition, or fix product/ops?

## Framework

Validate -> Decompose (mix vs rate) -> Hypothesis tree -> Size each driver into a waterfall -> Recommend with holdouts and guardrails.

## Steps

### Step 1 - Is it real?

- **Right-censoring:** recent cohorts have not had the full 90 days, so they look worse. Compare matured cohorts only.
- **Definition or ID changes:** login changes, guest checkout, merged IDs can split one customer into two and fake lower repeat.
- **Calendar:** festivals and long weekends shift between years. Compare like-for-like windows.
- **Pipeline:** reconcile server-side bookings with client events.

### Step 2 - Mix vs rate decomposition (key analytical move)

A blended repeat rate can fall with no segment getting worse if the mix shifted toward low-repeat segments (Simpson's paradox). Total change = mix effect + rate effect, where mix = sum of (change in share x base-year rate) and rate = sum of (new share x change in rate).

### Worked example (made-up numbers)

| | Last year | This year |
|---|---|---|
| Organic: share / repeat | 60% / 30% | 40% / 29% |
| Paid: share / repeat | 40% / 20% | 60% / 19% |
| Blended repeat | 26% | 23% |

Mix effect = (-20% x 30) + (+20% x 20) = -2.0 pts. Rate effect = (0.4 x -1) + (0.6 x -1) = -1.0 pt. Two-thirds of the drop is acquisition mix, not worse behaviour; the fix is acquisition targeting, not CLM.

Cut by: acquisition channel, vintage, new vs reactivated, route type (commuter / long-haul / festival), tier-1 vs tier-2, platform, first-trip price band, operator type (private vs state RTC).

### Step 3 - Hypothesis tree

- **Acquisition quality:** more paid or discount-led, low-intent users who are one-and-done.
- **First-trip experience:** cancellations, delays, refund time, no-shows. Link first-trip NPS/CSAT to repeat.
- **Supply and product:** fewer operators or routes on key corridors, seat availability, search or filter changes, app release issues.
- **Substitutes:** rail, flights, other bus apps, operator direct apps on specific corridors.
- **CLM changes:** fewer sends, frequency caps, channel shift (WhatsApp opt-ins falling), template changes.
- **Pricing and incentives:** coupons pulled back, or heavy first-order discounts creating discount hunters.
- **External:** fuel and fares, strikes, regulation, weather.

### Step 4 - Validate and size each hypothesis

| Hypothesis | Confirming evidence | How to size |
|---|---|---|
| Acquisition mix | Mix effect large; paid-cohort repeat consistently lower | Mix-effect points from Step 2 |

| Hypothesis | Confirming evidence | How to size |
|---|---|---|
| First-trip experience | Users with a cancellation or slow refund repeat far less; incidence up YoY | Incidence x repeat gap = points lost |
| Supply | Drop concentrated on corridors with falling inventory | Corridor repeat vs supply index |
| CLM | Repeat fell where contact rate fell; holdout lift lower | Contacted vs historical |
| Competition | Drop on rail-heavy or flight-heavy corridors | Corridor overlap analysis |

Goal: a waterfall that explains the 3 points, e.g. mix -1.8, first-trip issues -0.7, CLM -0.3, unexplained -0.2 (illustrative). State anything unexplained openly. Causal caution: 'bad first trip users repeat less' is correlational; control for route, price and channel, or use DML or matching.

### Step 5 - Recommend

| Finding | Action |
|---|---|
| Mix shift to low-repeat channels | Judge channels on repeat-adjusted LTV/CAC; cut or reprice worst paid segments |
| First-trip friction | Proactive comms and compensation on disruption; ops owns refund-time SLAs |
| CLM gap | Rebuild next-trip timing and return-leg triggers; target by uplift |
| Corridor competition | Corridor-specific offers or supply pushes |

Measure: primary metric = 90-day repeat of matured cohorts; holdout on each intervention; guardrails = unsubscribe rate, NPS, margin.

## Pitfalls

- Reading the blended number without decomposing.
- Using immature cohorts.
- Treating correlation (bad trip vs low repeat) as proven cause.
- Jumping to CLM fixes when the cause is acquisition mix.

## Resume hook

Meta holdout (separating real from attributed), churn model with test/control, DML and RCA work (50% faster RCA).

## Likely probes

- **How do you tell mix from rate quickly?** Run the decomposition; one query once segments are defined (see SQL section).
- **Which hypothesis first?** Mix and censoring (cheap, can invalidate everything); then first-trip experience, the likeliest real driver in travel.
- **How do you prove first-trip experience causes lower repeat?** Control confounders (matching or DML), find a quasi-random event such as operator cancellations, or run a compensation holdout.
- **3 points is small; why care?** Convert to money: points x base x trips x margin per booking.

## Spoken answer (about 2 minutes)

*First I confirm the drop is real: matured cohorts only, same definition both years, tracking checked. Then I decompose into mix versus rate, because a blended repeat rate can fall with no segment getting worse if acquisition shifted toward low-intent paid users. If it is mostly mix, the fix is acquisition targeting and repeat-adjusted LTV to CAC. If it is rate, I go down a hypothesis tree: first-trip experience, supply on key corridors, CLM changes, competition from rail. I size each into a waterfall that explains the 3 points, then recommend actions, each with a holdout and NPS and unsubscribes as guardrails.*

# Case 2: Design a CLM program to lift repeat

*Prompt: Design a CLM program to lift repeat. How do you pick timing, channel and message?*

## Step 0 - Clarify (say these out loud)

- Goal: repeat rate, repeat revenue, or cross-sell? The primary metric drives design.
- Constraints: channels available, opt-in base, cost per message, discount budget.
- Scope: all users or one LoB / geography first?
- Current state: what already runs and is there a holdout?

## Framework

Three decisions per user - Who, When, What (channel + message + incentive) - then Measure. Two-stage targeting: eligibility (propensity or next-trip window) then prioritization (uplift per rupee).

## Steps

### Step 1 - Frame

Bus is episodic, so 'send more campaigns' is the wrong instinct. CLM must decide who to contact (and who to leave alone), when a next trip becomes likely, and what channel, message and incentive to use.

### Step 2 - Segment by trip behaviour, not just recency

| Segment | Behaviour | CLM lever |
|---|---|---|
| Commuters | Same route weekly or monthly | Route and day reminders, price alerts, convenience features |
| Return-leg | Booked one way, not yet returned | Return-trip nudge timed to the trip |
| Festival / seasonal | Travel at Diwali, Pongal, summer | Pre-festival 'book early' reminders by annual cohort |
| Leisure / one-off | Occasional, destination-driven | Destination content and cross-sell (hotel) |
| Lapsed | Past their typical gap | Route-specific win-back, only if persuadable |
| First-timers | One booking | Post-trip experience follow-up; the most valuable repeat window |

Build segments from data (route concentration, gap regularity), not assumptions.

### Step 3 - Timing: model it, do not guess

- Estimate each segment's inter-booking gap distribution; trigger when the next booking is becoming likely but has not happened (e.g. median or 70th percentile of the gap).
- Better: a survival / hazard model of days-to-next-booking with route, past gap and seasonality as features; output a predicted next-trip window per user.
- Event triggers beat calendar blasts: pre-trip (T-1; boarding info is high engagement and a cross-sell slot), post-trip (T+1 to T+2: rating, return-leg offer), price drop or availability on the usual route, festival countdown.

### Step 4 - Channel selection

| Channel | Cost | Best for |
|---|---|---|
| Push | Near zero; opted-in app users | Timely triggers, reminders |
| WhatsApp | Per-message fee; high engagement | High-value moments: return-leg, win-back |
| SMS | Per-message; reaches everyone | Transactional; users without the app |
| Email | Cheapest; low engagement | Content, receipts |

Rule: cheapest channel that delivers incremental lift. Run channel tests; later move to a contextual bandit or per-user uplift by channel. Cap frequency across all channels; fatigue is the main way CLM fails.

### Step 5 - Message and incentive

- Lead with non-discount value: reminders, convenience, price alerts, 'your usual 9pm route is open'.
- Incentive only where uplift exists. Discounting always-repeaters wastes money and trains coupon waiting. Train the uplift model on your holdout data.

- A/B test message variants (urgency, route personalization, social proof) and test discount levels to find the cheapest that moves behaviour.

### Step 6 - Targeting logic (two-stage)

Stage 1 eligibility: propensity to lapse or predicted next-trip window. Stage 2 prioritization: uplift score, ranked by expected incremental margin per rupee = (incremental booking value x margin) - comms cost - incentive. Same structure as your festive churn story.

### Step 7 - Measurement

- Persistent global holdout (5-10% of users get no CLM) for whole-program incremental repeat; campaign-level holdouts inside each journey.
- Primary metric: incremental repeat bookings or 90-day repeat; incremental margin. ROI = (incremental margin - comms cost - incentive cost) / cost.
- Guardrails: unsubscribe and opt-out rate, complaints, NPS, uninstalls, margin per booking.
- Pull-forward check: did we create a booking or only move one earlier? Look for a dip afterwards. Analyse intent-to-treat; do not judge on opens or clicks.

### Step 8 - Rollout

| Phase | Action |
|---|---|
| Weeks 1-4 | Segment, find gap distributions, set up global holdout, launch 2-3 triggers (return-leg, post-trip, lapsed route) |
| Months 2-3 | Timing model, channel tests, begin uplift targeting |
| Months 3-6 | Cross-sell journeys (Bus to Hotel/Rail), bandit optimization, automation |

Illustrative sizing (made-up): 1M eligible users a month, 0.8% incremental repeat, Rs 100 margin, Rs 0.30 comms cost per user: incremental margin = 8,000 x 100 = Rs 8L vs cost Rs 3L, ROI about 2.7x. Shows you size before building.

## Pitfalls

- Blasting all users; ignoring frequency caps.
- Discounting persuadables and always-repeaters alike.
- Judging on opens and clicks instead of incremental bookings.
- No persistent holdout, so cumulative fatigue is invisible.

## Resume hook

Festive churn model with test/control; 100K+ customers retained monthly; instant Supercoins credit test (10 bps conversion uplift); Loyalty Handbook (repeatable playbooks).

## Likely probes

- **Frequency?** Test caps across cohorts; best cap is where marginal lift equals marginal fatigue cost.
- **Users who would repeat anyway?** Uplift model plus holdouts; skip sure repeaters unless contact is free and non-incentivized.
- **Long gaps between bookings?** Long window (90+ days), persistent holdout, leading indicators (opens, searches) while the outcome matures.
- **A channel's opt-in is falling?** Check consent flow vs fatigue, test value-first copy, shift volume to stable channels.
- **Different from Flipkart?** Same mechanics; purchase frequency is lower so timing and route context matter more than volume.

## Spoken answer (about 2 minutes)

*I would build CLM as three decisions per user: who, when, and what. Segment by trip behaviour: commuters, return-leg, festival travellers, one-off leisure and lapsed. For timing I model each segment's booking gap and trigger when the next booking becomes likely, plus event triggers like post-trip and return-leg. For channel, the cheapest that delivers incremental lift, with a frequency cap across channels. For message, value first, incentives only where an uplift model says behaviour changes. I measure with a persistent holdout, counting incremental repeat and margin, with unsubscribes and NPS as guardrails.*

# Case 3: CAC is rising and budget is flat

*Prompt: Our CAC is rising and budget is flat. What do you do?*

## Step 0 - Clarify (say these out loud)

- How is CAC defined: blended or paid only, new customers only, which attribution?
- Since when, and which channels or geographies?
- Is quality (repeat, LTV) holding?
- Any tracking or attribution change recently?

## Framework

Define -> Decompose (mix vs rate, then funnel math) -> Check attribution -> Act in three places (cut, reallocate, convert better) -> Guardrails.

## Steps

### Step 1 - Define and decompose

- CAC = spend / new customers. Decompose blended CAC change by channel: mix effect (spend shifted to costlier channels) vs rate effect (each channel got costlier).
- Within a channel: CAC = CPM / (1000 x CTR x conversion rate). Find which term moved: auction cost (CPM), creative (CTR), or landing and funnel (conversion).

### Step 2 - Is it real?

Check for tracking changes, attribution-window changes, double counting, and new-vs-reactivated misclassification that can inflate or deflate CAC.

### Step 3 - Incremental lens

Reported CAC is attributed. A channel with low reported CAC may mostly harvest existing intent (branded search, retargeting). Use holdouts or geo tests to estimate incremental CAC for the top-spend channels.

### Step 4 - Act in three places

- **Cut:** pause low-incrementality spend (brand-keyword overlap with organic, retargeting of likely converters).
- **Reallocate:** build response curves; move budget until marginal ROAS equalizes across channels.
- **Convert better:** if CAC rose because conversion fell, fix funnel friction instead of buying more traffic.
- **Shift to owned channels:** CLM and referral have lower marginal cost.

### Step 5 - Judge on value, not CAC alone

Use repeat-adjusted LTV/CAC and payback period. A cheap acquirer of one-time users can be worse than a pricier channel that brings repeaters.

## Recommendations / what to do differently

- Mix-vs-rate readout in the first week; holdouts on top 2-3 channels in weeks 2-4.
- Reallocation plan with marginal ROAS per channel.
- A CAC guardrail paired with a repeat-rate guardrail so efficiency does not buy low-quality users.

## Pitfalls (shows depth)

- Optimizing average ROAS instead of marginal.
- Cutting a channel on reported CAC when its incremental share is unknown.
- Cheap CAC from discount-led acquisition that never repeats.

## Resume hook

CAC reduced to below Rs 250 (30% reduction) while acquiring about 300K incremental customers a month; Meta holdout. Be ready to state how CAC was defined and what happened to repeat.

## Likely probes

- **How would you find channel saturation?** Spend-response curves from historical variation, geo spend tests, or MMM.
- **What if finance wants CAC below a hard cap?** Show the cap's effect on volume and repeat; propose a payback-period cap instead.

## Spoken answer (about 2 minutes)

*First I find why CAC rose: mix versus rate, then within channels the CPM, click-through and conversion terms. I check it is not a tracking artifact. Then I act three ways: cut the least incremental spend, reallocate to equalize marginal ROAS, and fix funnel conversion. I judge channels on repeat-adjusted LTV to CAC, since cheap one-time users are not cheap.*

# Case 4: Is this channel actually working? (incrementality and attribution)

*Prompt: How do you know if a marketing channel is working? Where has attribution misled you?*

## Step 0 - Clarify (say these out loud)

- Which channel and what is the decision (scale, hold, cut)?
- Can we randomize (addressable channel) or only by geography/time?
- Time horizon: first booking or repeat value?

## Framework

Evidence ladder: randomized holdout > geo / quasi-experiment (DiD, matched geos) > MMM > multi-touch attribution > last-touch. Use the strongest feasible rung and calibrate the weaker ones to it.

## Steps

### Step 1 - Why last-touch misleads

- Credits people who already planned to buy (existing intent), especially bottom-funnel: brand search and retargeting.
- Under-credits upper funnel and brand.
- View-through counting inflates credit for impressions.
- Averages hide variation: some business units or categories work, others do not.

### Step 2 - Design the holdout

- Randomize at user level into exposed and holdout; hold out a fraction large enough for the minimum detectable effect on the primary metric.
- Size with a power calculation across candidate windows (e.g. 2 vs 3 weeks); fix duration and analysis date upfront; no peeking.
- SRM check: confirm the split ratio is as designed.
- For non-addressable channels (TV, OOH), use geo experiments with matched markets and difference-in-differences.

### Step 3 - Read out

- Incremental customers, incremental revenue and margin, incremental cost per customer, ROI on incremental revenue.
- Report weekly as data arrives; customer-level significance usually arrives before revenue-level (revenue is higher variance).
- Split by business unit or segment because each owns spend, and drill into categories behind weak results.
- Check quality: 90-day repeat and LTV of incremental customers, not only first-booking CAC.

### Step 4 - Your Meta example (use this)

8% holdout sized from minimum detectable effect on customer conversion; weekly readouts; significance on incremental customers in about 2 weeks and on incremental GMV about 1 week later; split by business unit, then super-category for weak BUs; finding: channel healthy overall so budgets were scaled, with targeting corrections for weak BUs. Do not quote an attributed-vs-incremental gap number unless you remember it; say 'directionally different'.

### Step 5 - MMM as the ongoing layer

Use tests as ground truth, then MMM for continuous allocation (adstock, saturation, controls), calibrated to test results. Tests give one channel at one time; MMM covers the portfolio over time.

## Recommendations / what to do differently

- Run holdouts on the top-spend addressable channels first; geo tests for brand channels.
- Replace last-touch ROAS in reviews with incremental ROAS where tests exist.
- Reallocate on marginal incremental ROAS.

## Pitfalls (shows depth)

- Treating a holdout result as permanent (effects drift).
- Contamination if holdout users see the ads through other channels.
- Judging brand on short windows.

## Resume hook

Meta incrementality test (8% holdout); Flipkart ROAS context: Rs 50 Cr monthly spend, Rs 600 Cr last-touch revenue.

## Likely probes

- **Why not just use MTA?** It still relies on observed touches and cannot see the counterfactual.
- **What is a PSA / ghost-ad design?** Show control users a neutral ad (or log would-have-been-shown) to isolate exposure from targeting.

## Spoken answer (about 2 minutes)

*Working means incremental, not attributed. I keep a randomized slice unexposed, size it from the minimum detectable effect, and compare conversion and margin against the exposed group. In our Meta test, the incremental picture differed by business unit in ways last-touch hid, which changed where budget went. For channels I cannot randomize I use geo tests, and I calibrate MMM to those results.*

# Case 5: Portfolio decisions: acquisition vs retention, brand vs performance

*Prompt: How do you balance acquisition against retention spend? How do you decide brand vs performance?*

## Step 0 - Clarify (say these out loud)

- What is the budget, horizon and payback target?
- Current LTV/CAC by cohort and channel?
- Do we have MMM or test evidence for brand effects?

## Framework

Put everything in one currency (12-month contribution margin per rupee) and compare marginal returns across options; add time horizon and uncertainty for brand.

## Steps

### Part A - Acquisition vs retention

- Acquisition: incremental CAC vs expected LTV of the new customer.
- Retention: incremental margin from CLM (via holdouts) vs program cost.
- Shift money to the higher marginal LTV-to-cost until they equalize.
- Retention is cheaper per unit but capped by base size; acquisition feeds the base.
- Interaction: poor retention raises effective acquisition cost (leaky bucket). Churn sets the true cost of a customer.
- Segment retention spend by uplift; skip sure repeaters and lost causes.
- redBus angle: bus is episodic, so repeat is probably under-invested relative to payoff; verify with data.

### Part B - Brand vs performance

- Brand works slowly and shows up as lower acquisition cost and higher organic share; judging it on last-touch makes it always lose.
- Evidence: MMM (adstock, saturation) calibrated with geo tests; brand trackers (awareness, consideration); trend in organic and direct traffic.
- Start from a benchmark split, then adjust on MMM marginal returns; fund brand until its marginal contribution falls below performance's.
- Judge brand over 6-12 months; move in steps (increase brand in some markets, watch organic bookings and performance efficiency).
- Performance harvests demand brand creates; cutting brand tends to raise performance CAC with a lag.
- Be honest about uncertainty: brand effects are noisy.

## Recommendations / what to do differently

- A single portfolio view with LTV/CAC and payback by cohort.
- Stepwise brand tests by market rather than one big shift.

## Pitfalls (shows depth)

- Comparing average, not marginal, returns.
- Judging brand on weekly performance metrics.

## Resume hook

Churn model with randomized at-risk test/control; Meta holdout; MMM listed under methodologies (state clearly if you have not built one).

## Likely probes

- **Rule of thumb you would check?** LTV/CAC and payback by cohort.
- **Have you built an MMM?** No full MMM at team level; ran incrementality tests that calibrate it.

### Spoken answer (about 2 minutes)

*I treat it as one portfolio and compare marginal returns in a common currency: contribution margin per rupee over a fixed horizon. Retention is cheaper per unit but capped by base size, and weak retention makes acquisition costlier, so the two interact. For brand I would not use last-touch; I would use MMM calibrated by geo tests plus leading indicators, move in steps, and judge over 6-12 months.*

# Case 6: Opportunity sizing and new segments

*Prompt: Size the Rail cross-sell opportunity for redBus users. Then: identify a new high-growth segment.*

## Step 0 - Clarify (say these out loud)

- Time horizon (12 months?) and metric (bookings, GMV or margin)?
- Which users are in scope: active in last 12 months?
- Incremental to what: is rail demand additive or does it cannibalize bus?

## Framework

Bottom-up funnel for the base case, top-down cross-check, low/base/high range, sensitivity on the 2-3 assumptions that matter, and a validation plan (pilot).

## Steps

### Step 1 - Bottom-up funnel

Opportunity = active bus users x share on routes where rail is relevant x share with a rail need in the horizon x cross-sell conversion x margin per booking.

### Illustrative numbers (made-up)

8M active users x 35% rail-relevant x 20% with rail intent x 10% conversion = 56,000 bookings; x Rs 60 margin = about Rs 33.6 lakh. Show low/base/high by varying intent and conversion.

### Step 2 - Top-down cross-check

Market size of overlapping corridors x achievable share. If bottom-up and top-down differ by more than about 2x, find which assumption is off.

### Step 3 - Net, not gross

Subtract cannibalization (users who would have booked a bus) and overlap with existing rail bookers. Report net incremental margin to the group, not only rail GMV.

### Step 4 - Sensitivity and validation

Rank assumptions by impact on the answer (usually intent and conversion). Run a small pilot or holdout to replace the weakest assumption with data before scaling.

### Part B - New segments

- Cluster users on behaviour (route concentration, trip frequency, lead time, price sensitivity, cross-product use), not demographics alone.
- Score each segment on size, repeat rate, CAC, margin and growth.
- Examples to test: tier-2/3 city travellers, festival-only travellers, commuters, women travellers, cross-LoB users.
- Validate with a small pilot before scaling; check the segment is reachable and the economics hold after acquisition cost.

## Recommendations / what to do differently

- Present range plus the 2-3 assumptions that move it most.
- Recommend a pilot to firm up the weakest assumption.
- Prioritize LoBs/segments by net margin per effort.

## Pitfalls (shows depth)

- Single-point estimates with false precision.
- Ignoring cannibalization.
- Segments defined by demographics with no behavioural difference.

## Resume hook

Cohort deep dives (MPC profiling, funnel transitions); AOP sizing work at Flipkart (2025/26 AOP).

## Likely probes

- **How accurate is this?** Directional; the pilot replaces the shakiest assumption.
- **Which assumption would you check first?** The one with the biggest swing in the sensitivity table.

## Spoken answer (about 2 minutes)

*I size bottom-up: active users, times the share on rail-relevant routes, times rail intent, times conversion, times margin, then cross-check top-down. I net out cannibalization, show a low, base and high range, and name the two assumptions that move it most. Then I propose a small pilot to replace the weakest assumption with data before scaling.*

# Case 7: Bus to Hotel cross-sell: timing and proof

*Prompt: Recommend the best time to cross-sell a hotel to a bus customer. How do you prove it worked?*

## Step 0 - Clarify (say these out loud)

- Goal: hotel attach rate, incremental hotel GMV, or group-level margin?
- Which bus trips are leisure vs commute (cross-sell fit)?
- Channels and discount budget?

## Framework

Fit (who) -> Timing candidates -> Randomized timing test with holdout -> Measure incremental attach and group margin -> Guardrails.

## Steps

### Step 1 - Fit: who is a hotel candidate?

Use destination, trip purpose proxies and history. A commuter returning home is rarely a hotel buyer; a tier-1 to leisure destination weekend trip may be. Build a hotel propensity score from destination type, stay length, lead time and past hotel use.

### Step 2 - Timing candidates

At booking confirmation; T-7 to T-3 days before travel; T-1 (boarding info); post-arrival (for the return or extended stay); on return-leg booking. Lead time matters: late hotel bookings convert at a different price point.

### Step 3 - Experiment

- Arms: each timing option plus a no-offer holdout; randomize at user level among eligible users.
- Primary: incremental hotel bookings per eligible user and incremental group margin.
- Guardrails: bus cancellation rate, unsubscribe rate, NPS, bus-booking conversion (do not clutter the bus flow).
- Read at intent-to-treat; check pull-forward and cannibalization of organic hotel bookings.

### Step 4 - Scale

Move to per-user timing using a model (propensity + uplift), or a bandit across timing slots, once the test shows which windows work for which segment.

## Recommendations / what to do differently

- Launch the best timing per segment; keep a persistent holdout.
- Report cross-sell economics at group level, not per-LoB silo.

## Pitfalls (shows depth)

- Offers that interrupt the core bus journey.
- Counting hotel bookings that would have happened anyway.
- Ignoring identity-resolution gaps across LoBs (see unified profile).

## Resume hook

Loyalty and Supercoins experiments (instant credit test); Netflix partnership instrumentation (check how cross-product it was).

## Likely probes

- **How do you build the unified profile?** Identity resolution on phone/email/device, consent management, one customer ID across LoBs, trip-level features.
- **What if the groups own different P&Ls?** Agree a group-level margin metric before launch.

## Spoken answer (about 2 minutes)

*I first decide who is a hotel candidate using destination and trip type, since a commuter going home is not. Then I test timing options such as booking confirmation, a few days before, and the day before, against a no-offer holdout, measuring incremental hotel bookings and group margin, with bus cancellations, unsubscribes and NPS*

*as guardrails. Once I know which windows work for which segment, I move to per-user timing.*

# Case 8: NPS and feedback to predict churn

*Prompt: NPS dropped 5 points. How do you use NPS and feedback to predict churn and improve engagement?*

## Step 0 - Clarify (say these out loud)

- Which segment dropped (route, operator, geography, new vs repeat)?
- Is it survey response mix or a real shift?
- Link between NPS and repeat today?

## Framework

Validate the signal -> Link NPS to repeat (churn-risk curve) -> Mine verbatims -> Extend beyond responders -> Close the loop -> Measure with holdouts.

## Steps

### Step 1 - Validate

Check response rate and mix change (detractors answer more; survey trigger changes can move NPS with no real change). Compare like-for-like segments.

### Step 2 - Link NPS to repeat

Compute subsequent repeat rate by NPS band (promoter / passive / detractor), controlling for route, price and channel. This gives the churn-risk uplift of a detractor and turns NPS into money.

### Step 3 - Mine verbatims

Classify comments with an LLM into topics (operator behaviour, delay, cancellation, refund, boarding, pricing, app). Validate on a human-labelled sample; track topic share and topic-level NPS.

### Step 4 - Extend beyond responders

Only a minority respond. Build a churn model using operational proxies available for everyone (cancellation, delay, refund time, rating, support contact) with NPS and topics as extra features for responders. Treat NPS as a label for a satisfaction model that scores non-responders.

### Step 5 - Close the loop

Detractor recovery (outreach, compensation, ops escalation); route and operator fixes by topic with owners; feed insights to supply and ops, not only marketing.

### Step 6 - Measure

Randomize recovery outreach with a holdout; measure incremental repeat and NPS recovery; guardrail on compensation cost per saved customer.

## Recommendations / what to do differently

- Topic-ranked list of fixable causes with sizing (points of repeat at stake).
- Recovery program with a holdout.
- Operator scorecards using NPS and cancellation rates.

## Pitfalls (shows depth)

- Treating NPS as causal.
- Over-weighting detractors' loud topics over frequent ones.
- Compensation to users who would have repeated anyway (use uplift).

## Resume hook

Churn model work; RCA on journey friction (Rs 1 Cr monthly savings); Claude skills for classification and RCA.

## Likely probes

- **How do you know the model is not just detecting trip failures?** That is fine if actionable; check lead time and whether intervention changes outcomes.
- **LLM classification accuracy?** Human-labelled validation sample, confusion matrix per topic, periodic drift checks.

## Spoken answer (about 2 minutes)

*First I check whether the drop is a response-mix artifact. Then I link NPS bands to subsequent repeat to get the churn-risk uplift of a detractor. I classify verbatims into topics with an LLM validated on a human sample, extend scoring to non-responders using operational proxies, and close the loop with detractor recovery tested against a holdout and fixes owned by ops and supply.*

# Case 9: Cohort-level predictions and an AI-led analytics team of two

*Prompt: Move from generic forecasts to cohort-level predictions. How would you use AI deeply with a team of 2?*

## Step 0 - Clarify (say these out loud)

- What decisions need forecasts (marketing spend, inventory, comms volume)?
- Granularity needed: route, city, cohort, week?
- What tooling and data access exists?

## Framework

Decisions first -> Hierarchical cohort forecast -> Accuracy by cohort -> AI automation layer with guardrails -> Operating model for a team of 2.

## Steps

### Step 1 - Cohort-level forecasting

- Forecast by cohort x route cluster x season using hierarchical methods (bottom-up with reconciliation to the total).
- Features: tenure, booking gap, festival calendar, price, supply index, marketing exposure.
- Track error by cohort (not only overall); use forecasts for budget pacing, comms volume and supply asks.

### Step 2 - AI automation layer (your real examples)

- Automated RCA playbooks and SQL generators (your Claude skills; 50% faster RCA).
- Weekly insight digests with anomalies and recommended actions.
- LLM classification of NPS verbatims and support tickets.
- Agentic alerts on metric deviations with first-pass diagnosis.

### Step 3 - Guardrails

Metric glossary and certified definitions as the source of truth for AI; validation checks (reconcile totals, test queries on known answers); human sign-off before decisions; log prompts and outputs; never let AI invent numbers.

### Step 4 - Operating model for a team of 2

- Triage by decision value; protect 50% of time for proactive growth questions, not requests.
- Self-serve templates and a handbook (your Loyalty Handbook) so stakeholders answer routine questions.
- Hands-on yourself: you build the first pass, then productize.
- Weekly cadence with Brand, CLM and Data Science: one page of 'what to do differently'.

## Recommendations / what to do differently

- Cohort forecast v1 in 60 days; AI RCA and digest pipeline in 30-60 days.
- A metric glossary before automation.

## Pitfalls (shows depth)

- Automating bad definitions.
- Spending the team's time on reporting.
- Black-box forecasts with no cohort-level accuracy tracking.

## Resume hook

Claude skills adopted across the Loyalty pod; WBR insight generators; Loyalty Handbook; team of 3 analysts.

## Likely probes

- **How do you make sure AI-generated insights are right?** Certified metrics, reconciliation checks, sampled human review, and tracing each number to a query.
- **Why a team of 2 is enough?** Leverage through automation and self-serve, and strict prioritization.

### Spoken answer (about 2 minutes)

*I would move from one generic forecast to cohort-by-route forecasts reconciled to the total, with accuracy tracked per cohort, and use them for spend pacing, comms volume and supply asks. With a team of two I would automate RCA and weekly digests using AI, with a certified metric glossary and validation checks so numbers are trustworthy, and I would stay hands-on to build the first version myself.*

# 5. Director-round question bank, answers and 30/60/90

## Likely themes

| Theme | Questions |
|---|---|
| Role fit | Why marketing analytics and why redBus? What does 'not a reporting role' mean to you? Why a team of 2 after managing 3? How hands-on will you be? |
| Marketing judgment | Is a channel working? CAC rising, budget flat? Where did attribution mislead? Acquisition vs retention? Brand vs performance? |
| Repeat, CLM, cross-sell | Repeat rate dropped; improve repeat for episodic products; cross-sell hotel or rail; channel and frequency; how to measure a campaign |
| Working with marketing | A time your analysis changed a budget; a marketer disagrees with your data; prioritizing Brand, CLM and Performance requests; presenting to non-technical leaders |
| AI-led | How have you used AI in analytics? What would you build here? How do you validate AI insights? |
| Tough probes | No travel or email-led growth experience; first 30/60/90; where do you disagree with how marketing teams use data? |

## Model answers to the five marketing-judgment questions

### 1. How do you know if a marketing channel is actually working?

*Working means incremental, not attributed. A channel can show great last-touch ROAS simply by reaching people who would have bought anyway. So I ask what would have happened without the spend. The cleanest answer is a randomized holdout: keep a slice unexposed and compare conversions and revenue. Then I look at incremental cost per customer and ROI on incremental revenue, not reported numbers.*

- Evidence ladder: holdout > geo/quasi > MMM > MTA > last-touch.
- Check quality too: 90-day repeat and LTV of incremental customers.
- Guardrails: margin per booking, cancellation and refund rate.
- Proof point: Meta incrementality test.

### 2. Our CAC is rising and budget is flat. What do you do?

*First I find why: mix versus rate, then the CPM, click and conversion terms. I confirm it is not an attribution artifact. Then I cut the least incremental spend, reallocate to equalize marginal ROAS, and improve post-click conversion. I judge channels on repeat-adjusted LTV to CAC, because a cheap acquirer of one-time users can be worse than a pricier one that brings repeaters.*

- Proof point: CAC below Rs 250 with about 300K incremental customers a month.

### 3. Last-touch vs incrementality: where has attribution misled you?

*In our Meta analysis. Last-touch credited Meta for conversions after exposure, many of which would have happened anyway. The holdout showed the true picture differed by business unit: some were healthy, some weak, which last-touch averages away. That changed where budget went: we scaled where incremental ROI was healthy and corrected targeting where it was not.*

**State only outcomes you know. If you do not remember the attributed-vs-incremental gap, say 'directionally different', not a number.**

### 4. How do you balance acquisition against retention spend?

*One portfolio, compared on marginal return: which next rupee produces more long-term value, a new customer or a repeat from an existing one? Acquisition: incremental CAC against LTV. Retention: incremental margin from CLM measured with holdouts against cost. I shift toward the higher marginal return until they equalize. For a product like bus, I would expect repeat to be underinvested relative to payoff, but I would verify.*

### 5. How would you decide brand vs performance?

*Performance is measurable short-term; brand works slowly and shows up as lower acquisition cost and higher organic share. I would not judge brand on last-touch. I would use MMM calibrated with geo tests, plus leading indicators like awareness and organic share, start from a benchmark and adjust on marginal returns, and move in*

*steps with tests. I would be honest that brand estimates are noisy.*

**Honesty note: MMM is a listed methodology but you said your team has not run one. Frame as 'here is my approach', not past experience.**

## First 30 / 60 / 90 days (draft)

| Window | Focus | Output |
|---|---|---|
| Days 1-30 | Learn the business: routes, seasonality, funnel, CLM calendar, data and tools. Audit metric definitions (repeat, CAC, active). Meet Brand, CLM, DS. | Metric glossary draft; repeat-rate decomposition (Case 1); top 3 growth hypotheses |
| Days 31-60 | Set up measurement: persistent CLM holdout, holdouts on top paid channels, first AI-based RCA and weekly digest. | First incrementality readouts; CLM trigger tests live |
| Days 61-90 | Move to decisions: reallocation plan on marginal ROAS, segment and opportunity sizing across LoBs, cross-sell pilot design. | Budget recommendations; opportunity-sizing deck; 6-month roadmap |

## Behavioural question starters (prepare a story for each)

- Analysis changed a budget or campaign decision (Meta holdout).
- You challenged an assumption and it changed a decision.
- A stakeholder disagreed with your numbers.
- Prioritizing with a small team.
- Raising a junior analyst's work.
- A result you got wrong and what you changed.

# 6. SQL toolkit for marketing analytics

Assumed schema (adapt names). Dialect: BigQuery-style standard SQL; for other warehouses adjust date functions.

| Table | Key columns |
|---|---|
| bookings | booking_id, user_id, booking_ts, travel_date, lob (bus/hotel/rail), route_id, gmv, margin, status (completed/cancelled) |
| users | user_id, signup_ts, acq_channel, acq_ts, city_tier |
| routes | route_id, origin, dest |
| comms_log | user_id, sent_ts, channel, campaign_id |
| assignment | user_id, assign_ts, group_flag ('test' / 'holdout'), campaign_id |
| marketing_spend | month, channel, spend |
| nps | user_id, booking_id, score, response_ts |

### SQL habits that show seniority

- Filter status = completed; decide how cancellations and refunds count.
- Exclude immature cohorts (right-censoring) when measuring windowed repeat.
- Watch join fan-out that double counts users or revenue.
- Use one canonical user ID; handle guest bookings and merged IDs.
- Be explicit on time zones and date boundaries.
- Run SRM and row-count sanity checks before reading results.

## Q1. 90-day repeat rate for matured cohorts (by cohort month and channel)

```
WITH completed AS (
  SELECT user_id, booking_ts
  FROM bookings
  WHERE status = 'completed' AND lob = 'bus'
),
first_b AS (
  SELECT user_id, MIN(booking_ts) AS first_ts
  FROM completed
  GROUP BY user_id
),
flags AS (
  SELECT f.user_id, f.first_ts, u.acq_channel,
         DATE_TRUNC(DATE(f.first_ts), MONTH) AS cohort_month,
         MAX(CASE WHEN c.booking_ts > f.first_ts
                   AND c.booking_ts <= TIMESTAMP_ADD(f.first_ts, INTERVAL 90 DAY)
                  THEN 1 ELSE 0 END) AS repeat_90
  FROM first_b f
  JOIN users u USING (user_id)
  LEFT JOIN completed c ON c.user_id = f.user_id
  GROUP BY f.user_id, f.first_ts, u.acq_channel
)
SELECT cohort_month, acq_channel,
       COUNT(*) AS users, AVG(repeat_90) AS repeat_rate_90d
FROM flags
WHERE first_ts <= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 90 DAY)  -- matured only
GROUP BY cohort_month, acq_channel
ORDER BY cohort_month, acq_channel;
```

## Q2. Mix vs rate decomposition of the YoY change (use flags from Q1)

```
-- flags_matured(user_id, cohort_year, acq_channel, repeat_90) derived from Q1
WITH seg AS (
  SELECT cohort_year, acq_channel, COUNT(*) AS n, AVG(repeat_90) AS r
  FROM flags_matured GROUP BY cohort_year, acq_channel
),
tot AS (SELECT cohort_year, SUM(n) AS N FROM seg GROUP BY cohort_year),
s AS (SELECT seg.*, n / N AS share FROM seg JOIN tot USING (cohort_year)),
cmp AS (
  SELECT a.acq_channel, a.share AS s0, b.share AS s1, a.r AS r0, b.r AS r1
  FROM s a JOIN s b ON a.acq_channel = b.acq_channel
  WHERE a.cohort_year = 2024 AND b.cohort_year = 2025    -- parameters
)
SELECT SUM((s1 - s0) * r0) AS mix_effect,
       SUM(s1 * (r1 - r0))  AS rate_effect,
       SUM(s1 * r1) - SUM(s0 * r0) AS total_change      -- = mix + rate exactly
FROM cmp;
```

## Q3. Inter-booking gap distribution by segment (feeds CLM timing)

```
WITH ordered AS (
  SELECT user_id, booking_ts,
         LAG(booking_ts) OVER (PARTITION BY user_id ORDER BY booking_ts) AS prev_ts
  FROM bookings WHERE status = 'completed' AND lob = 'bus'
),
gaps AS (
  SELECT user_id, DATE_DIFF(DATE(booking_ts), DATE(prev_ts), DAY) AS gap_days
  FROM ordered WHERE prev_ts IS NOT NULL
)
SELECT APPROX_QUANTILES(gap_days, 100)[OFFSET(50)] AS p50_days,
       APPROX_QUANTILES(gap_days, 100)[OFFSET(70)] AS p70_days,
       APPROX_QUANTILES(gap_days, 100)[OFFSET(90)] AS p90_days
FROM gaps;   -- join a user segment table and GROUP BY segment for per-segment timing
```

## Q4. Return-leg candidates: one-way travellers with no return booked (daily trigger list)

```
SELECT b.user_id, b.booking_id, b.travel_date
FROM bookings b
JOIN routes r ON r.route_id = b.route_id
LEFT JOIN (
  SELECT b2.user_id, b2.travel_date, r2.origin, r2.dest
  FROM bookings b2 JOIN routes r2 ON r2.route_id = b2.route_id
  WHERE b2.status = 'completed'
) ret
  ON ret.user_id = b.user_id
 AND ret.origin = r.dest AND ret.dest = r.origin
 AND ret.travel_date BETWEEN b.travel_date AND DATE_ADD(b.travel_date, INTERVAL 14 DAY)
WHERE b.status = 'completed' AND b.lob = 'bus'
  AND b.travel_date = DATE_ADD(CURRENT_DATE(), INTERVAL 1 DAY)   -- T-1 trigger
  AND ret.user_id IS NULL;
```

## Q5. Holdout readout (intent-to-treat) with SRM sanity check

```
WITH outcome AS (
  SELECT a.user_id, a.group_flag,
         MAX(CASE WHEN b.booking_ts BETWEEN a.assign_ts
                  AND TIMESTAMP_ADD(a.assign_ts, INTERVAL 30 DAY) THEN 1 ELSE 0 END) AS booked,
         COALESCE(SUM(CASE WHEN b.booking_ts BETWEEN a.assign_ts
                  AND TIMESTAMP_ADD(a.assign_ts, INTERVAL 30 DAY) THEN b.margin END), 0) AS margin_30d
  FROM assignment a
  LEFT JOIN bookings b ON b.user_id = a.user_id AND b.status = 'completed'
  WHERE a.campaign_id = 'CAMPAIGN_X'
  GROUP BY a.user_id, a.group_flag
)
SELECT group_flag,
       COUNT(*) AS users,                 -- SRM: compare ratio to designed split
       AVG(booked) AS conversion,
       AVG(margin_30d) AS margin_per_user
FROM outcome GROUP BY group_flag;
-- lift = test - holdout (absolute); test significance in Python (two-proportion z-test /
-- t-test on margin); incremental margin = lift x test users; compare to comms + incentive cost.
```

## Q6. CAC by channel with 90-day repeat (quality-adjusted view)

```
WITH new_cust AS (
  SELECT u.user_id, u.acq_channel, DATE_TRUNC(DATE(u.acq_ts), MONTH) AS month
  FROM users u
),
rep AS (
  SELECT n.user_id,
         MAX(CASE WHEN b.booking_ts > u.acq_ts
                  AND b.booking_ts <= TIMESTAMP_ADD(u.acq_ts, INTERVAL 90 DAY)
                  AND b.status = 'completed' THEN 1 ELSE 0 END) AS repeat_90
  FROM new_cust n JOIN users u USING (user_id)
  LEFT JOIN bookings b ON b.user_id = n.user_id
  GROUP BY n.user_id
),
agg AS (
  SELECT n.month, n.acq_channel, COUNT(*) AS new_customers, AVG(r.repeat_90) AS repeat_90
  FROM new_cust n JOIN rep r USING (user_id)
  GROUP BY n.month, n.acq_channel
)
SELECT a.month, a.acq_channel, a.new_customers,
       s.spend / a.new_customers AS cac,
       a.repeat_90,
       s.spend / NULLIF(a.new_customers * a.repeat_90, 0) AS cost_per_repeating_customer
FROM agg a JOIN marketing_spend s
  ON s.month = a.month AND s.channel = a.acq_channel;
-- reported CAC is attributed; pair with holdout-based incremental CAC for big channels.
```

## Q7. Monthly cohort retention matrix

```
WITH first_m AS (
  SELECT user_id, DATE_TRUNC(DATE(MIN(booking_ts)), MONTH) AS cohort_month
  FROM bookings WHERE status = 'completed' GROUP BY user_id
),
activity AS (
  SELECT DISTINCT user_id, DATE_TRUNC(DATE(booking_ts), MONTH) AS active_month
  FROM bookings WHERE status = 'completed'
)
SELECT f.cohort_month,
       DATE_DIFF(a.active_month, f.cohort_month, MONTH) AS months_since_first,
       COUNT(DISTINCT a.user_id) AS active_users
FROM first_m f JOIN activity a USING (user_id)
GROUP BY f.cohort_month, months_since_first
ORDER BY f.cohort_month, months_since_first;
-- divide by cohort size (months_since_first = 0) for retention %.
```

## Q8. NPS band vs subsequent 90-day repeat

```
WITH resp AS (
  SELECT user_id, booking_id, score, response_ts,
         CASE WHEN score >= 9 THEN 'promoter' WHEN score >= 7 THEN 'passive' ELSE 'detractor' END AS band
  FROM nps
),
fut AS (
  SELECT r.user_id, r.band,
         MAX(CASE WHEN b.booking_ts > r.response_ts
                  AND b.booking_ts <= TIMESTAMP_ADD(r.response_ts, INTERVAL 90 DAY)
                  AND b.status = 'completed' THEN 1 ELSE 0 END) AS repeat_90
  FROM resp r LEFT JOIN bookings b ON b.user_id = r.user_id
  GROUP BY r.user_id, r.band, r.response_ts
)
SELECT band, COUNT(*) AS responders, AVG(repeat_90) AS repeat_90
FROM fut
WHERE TRUE   -- add matured filter on response_ts
GROUP BY band;
-- correlation only: control for route, price and channel (matching/regression) before claiming cause.
```

# 7. Resume deep dive

Based on the 'Marketing Analytics & Data Science Leader' version. You also have a newer 'Lead Business Analyst (Data Science)' version; keep facts consistent across both.

## 7.1 Fix before the interview

| Issue | Risk |
|---|---|
| Titles and dates differ across versions (Lead Analyst from Jan 2022 vs Senior BA 2021-2024, Lead BA from Jan 2025) | MMT will cross-check LinkedIn and employment records |
| 'Senior BA Jan 2021 - Jan 2022' and 'Landmark 2021' overlap | Looks like a date error |
| Rs 6000 Cr GMV 'influenced' vs Rs 550 Cr annual GMV uplift on the other version | Know which is influenced and which is incremental |
| MDI PGDM (2026) appears on one version only | Use one consistent education line |
| Strongest 'AI-led' evidence (Claude skills, 50% faster RCA) is on the newer version only | Merge it into the version you submit |

## 7.2 Bullet-by-bullet probes

| Bullet | Likely probe | Prepare |
|---|---|---|
| Led analytics strategy for loyalty initiative, influencing Rs 6000 Cr GMV strategy | 'Influenced' is vague: what decision did you drive? How much is incremental? | One decision, your recommendation, outcome, honest influenced-vs-incremental split |
| Double ML to measure program impact, 4.5x lift in RPC | Why DML, not A/B or DiD? Treatment, outcome, confounders, first-stage models? Baseline? ATE or ATT? Confidence interval? Overlap and robustness checks? Opt-in selection bias? | DML in 3 sentences (residualize treatment and outcome on controls, regress residuals); name limits (unobserved intent); what the raw comparison showed vs DML-adjusted |
| Self-serve dashboards (Tableau, Looker) | JD says not a reporting role: which decision did a dashboard change? | Position as enablement; name one decision |
| A/B tests for new features | A test that surprised you? A no-ship recommendation? Sizing and peeking? | One story with the decision outcome |
| RCA on journey friction, Rs 1 Cr monthly savings | What friction? How was saving computed and verified? Recurring? Counterfactual? | Finding, fix, owner, savings math |
| Managed a team of analysts | Team size? Coaching example? How will you stay hands-on with a team of 2? | Exact size; one growth story |
| 300K incremental customers a month | Incremental by what method? Which channels? Spend vs ROAS? | Link to the Meta holdout; do not over-claim |
| Churn models retained 100K+ high-value customers monthly | 'Retained' vs 'predicted not to churn': what is your counterfactual? Label, features, metric, imbalance? | Use festive test/control story: 'we targeted; holdout showed X incremental' |
| XGBoost + SHAP driver models | Driver models vs causal impact; SHAP is association | Acknowledge it is explanatory, not causal |
| CAC down 30% to below Rs 250 | CAC definition? What drove it: mix, targeting, creative? Did repeat or quality fall? | Drivers and quality check |
| Landmark markdown/pricing; Impact Analytics promo optimization | How does pricing help marketing? | One line linking to elasticity and promo ROI |

## 7.3 Skills-section claims

| Claim | Probe | Safe answer |
|---|---|---|
| Marketing Mix Modeling | Walk me through an MMM you built | Team has not run a full MMM. You know the method and run the incrementality tests that calibrate it. Use any prior MMM if you have one |

| Claim | Probe | Safe answer |
|---|---|---|
| Attribution | Last-touch vs MTA vs MMM vs holdouts | Strong: Flipkart is last-touch based; Meta holdout |
| Marketo / HubSpot / Salesforce | What did you do in them? | Be exact on depth; say so if exposure only |
| LTV modeling | Method, horizon, decision use | Prepare one budget decision driven by LTV |
| Forecasting | Cohort-level predictions (JD) | Link to units prediction and impact forecasts |
| Cohort and funnel analysis | JD-relevant | A cohort finding that led to action |

## 7.4 Gaps an interviewer will spot

| JD need | On resume? | Prep |
|---|---|---|
| Travel / bus domain | No | One-line bridge on episodic repeat |
| Email/CLM channel optimization | Tools only | Prepare a real example |
| NPS / feedback to churn | No | Approach in Case 8, labelled as approach |
| Opportunity sizing | No | Rehearse Case 6 |
| Unified profile / cross-product | No | Check Netflix partnership relevance |

## 7.5 Questions generated from the resume

- Walk me through the DML analysis end to end. What would you do differently?
- How do you know the churn model's retained users were not going to stay anyway?
- Which of your numbers are incremental vs attributed? Defend one.
- Give an example where analysis, not reporting, changed a decision.
- How did you cut CAC 30% and what did it do to repeat?
- Where did you disagree with marketing or finance?
- What is different about repeat in bus?
- Which part of this role is new to you and how will you ramp up?

## Open items to tighten your stories

- 4.5x RPC: treatment (which program), baseline, main confounders?
- Rs 6000 Cr: what decision did you personally drive and what part is incremental?
- Marketo/HubSpot/Salesforce: hands-on or exposure?
- Which resume version is current for titles and dates?

# 8. 10-minute revision sheet

## Minute 0-2: Pitch

> *I lead loyalty analytics at Flipkart, 9 years in e-commerce. Three strengths: incrementality through holdouts and causal models, retention through churn and uplift targeting, and AI to speed insight generation. redBus repeat and cross-sell across Bus, Hotel and Rail is the same problem in a new domain.*

## Minute 2-4: Three flagship stories

| Story | Punchline |
|---|---|
| Meta holdout | 8% holdout sized by minimum detectable effect; BU-level results, then super-category; channel healthy, budgets scaled, weak BUs corrected |
| Festive churn model | LightGBM scored at-risk users; random test/control; measured reactivation, transactions, GMV. 'The holdout isolated true incremental lift.' Next step: uplift on top |
| DML impact (4.5x RPC) | No randomization, so residualize treatment and outcome on controls. Know baseline, confounders and selection-bias limits |

## Minute 4-6: Frameworks to recite

- **Case 1:** Real? -> Mix vs rate -> Hypotheses (acquisition quality, first trip, supply, CLM, competition) -> Waterfall -> Action + holdout.
- **Case 2:** Who (uplift) -> When (gap model, triggers) -> Channel (cheapest that lifts, frequency cap) -> Message (value first) -> Measure (global holdout, guardrails).
- **Case 3:** Define -> Mix vs rate and CPM/CTR/CVR -> Attribution check -> Cut, reallocate, convert -> LTV/CAC.
- **Case 4:** Evidence ladder -> Holdout design -> Readout by segment -> MMM calibrated to tests.
- **Case 6:** Bottom-up funnel -> Top-down cross-check -> Net of cannibalization -> Range + sensitivity -> Pilot.

## Minute 6-8: Traps to avoid

- Never say 'the model retained them'. Say 'we targeted, and the holdout showed X incremental'.
- Do not claim MMM, NPS or Marketo depth you do not have.
- 'Influenced' is not 'incremental'. Know the split for Rs 6000 Cr.
- SHAP explains; it does not prove causation.
- Lead with the decision, not the dashboard (the JD says not a reporting role).
- Keep dates and titles consistent across resume versions.
- Optimum is equal marginal returns, not 'split in the ratio of weights' (true only for log response).

## Minute 8-10: Questions to ask the Director

- Which 2-3 growth decisions do you wish had better data today?
- How is spend split across brand, performance and CLM, and how is it measured?
- What does success look like at 6 months?
- How does redBus approach Bus x Hotel x Rail cross-sell?
- How do Marketing Analytics and Data Science divide work today?